Simplicio Loop 4.0

Simulated benchmark projection

SIMULATED: no performance, cost, token or quality result in this report is a measured production claim.

Baseline release	3.38.5
Projection	4.0.0 - all issues in epic #801 completed
Baseline main SHA	16b8f94cf31a
Seed	20260728
Runs per scenario/workload	2500
Generated classification	SIMULATED

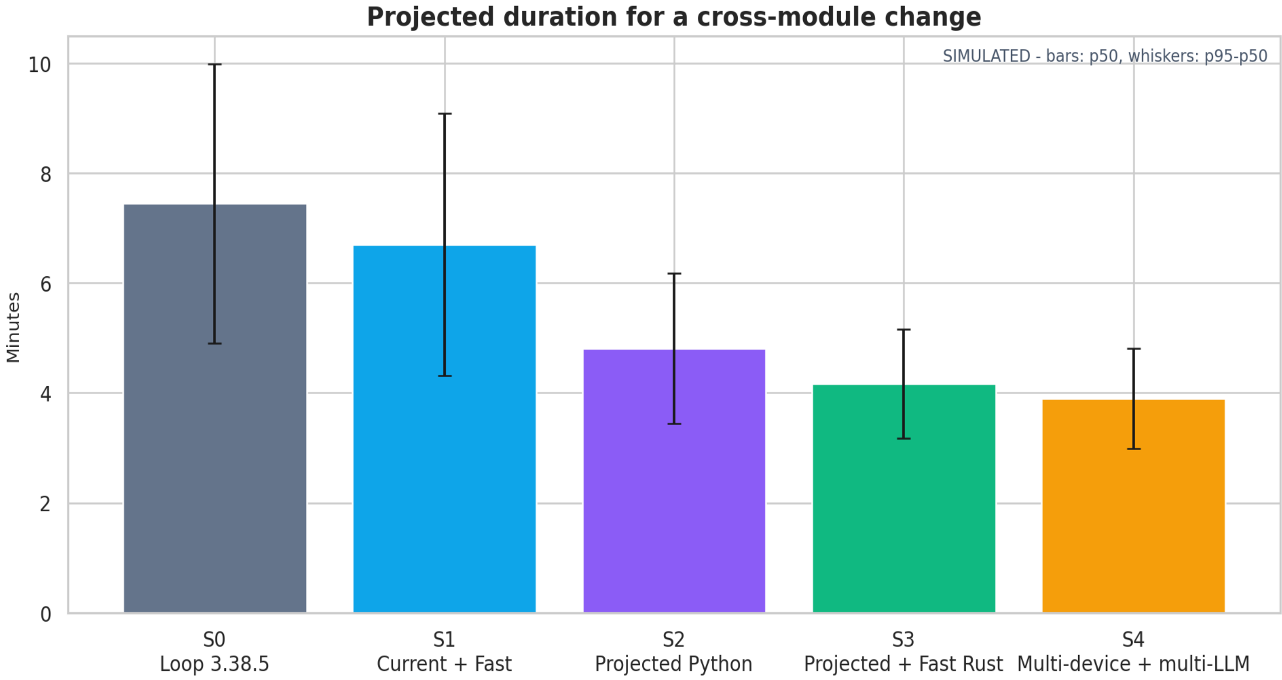
Purpose: estimate the potential shape of a release in which every child of epic #801 is complete, while preserving multi-device, multi-LLM, source reporting, recovery and evidence gates.

Executive projection

Workload	p50	p95	Token reduction	Completion
Mechanical change	69 s	81 s	72.5%	99.5%
Cross-module change	3.9 min	4.8 min	72.3%	96.3%
20 independent issues	3.2 min	4.2 min	72.5%	95.4%
100 issues with conflicts	9.3 min	12.9 min	72.5%	90.4%
Crash and recovery	3.7 min	5.4 min	72.3%	84.5%

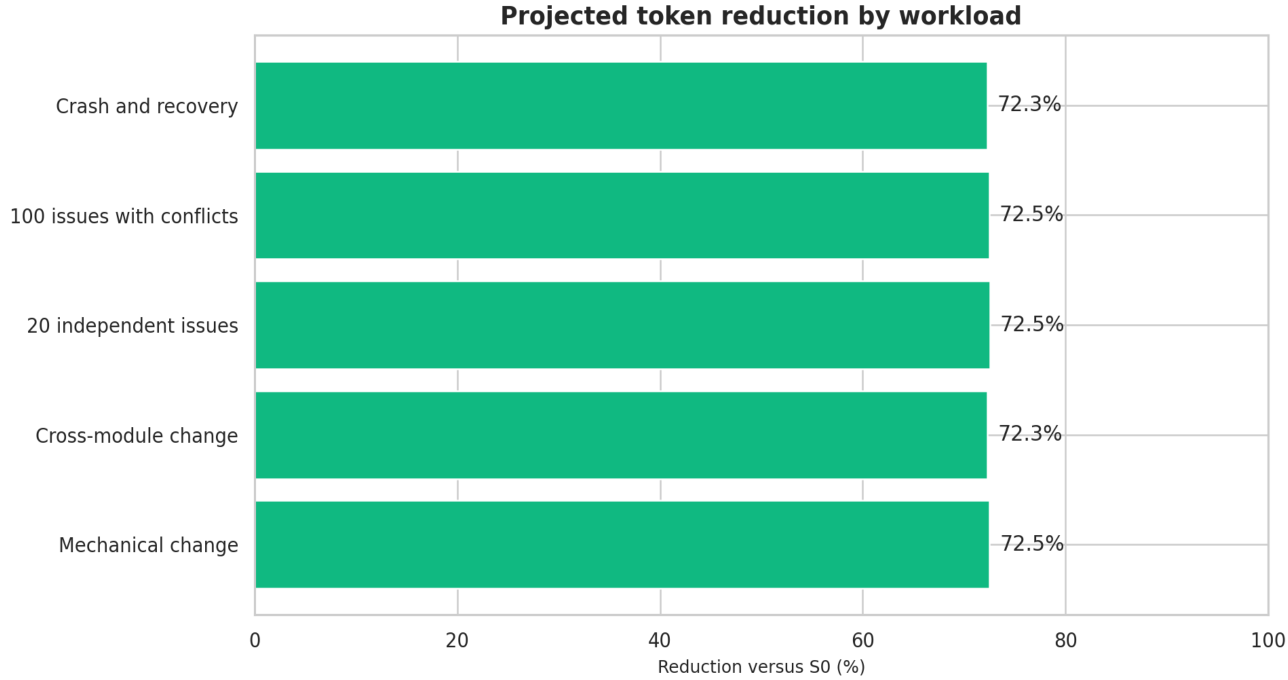
These values are hypotheses produced by the coefficients in assumptions.json. They indicate what to test, not what the current product has already achieved.

01 Duration Bar



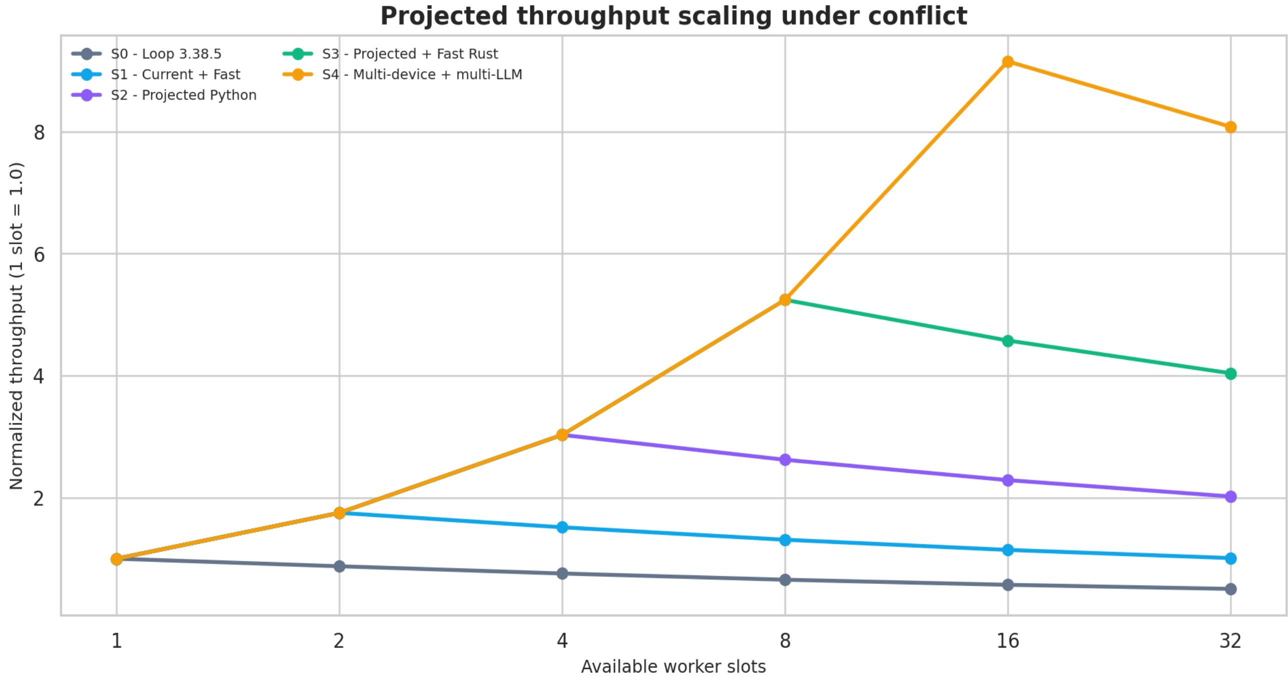
Duration combines phase-level uncertainty, queue jitter, retries and scenario-specific acceleration.

02 Token Reduction Horizontal



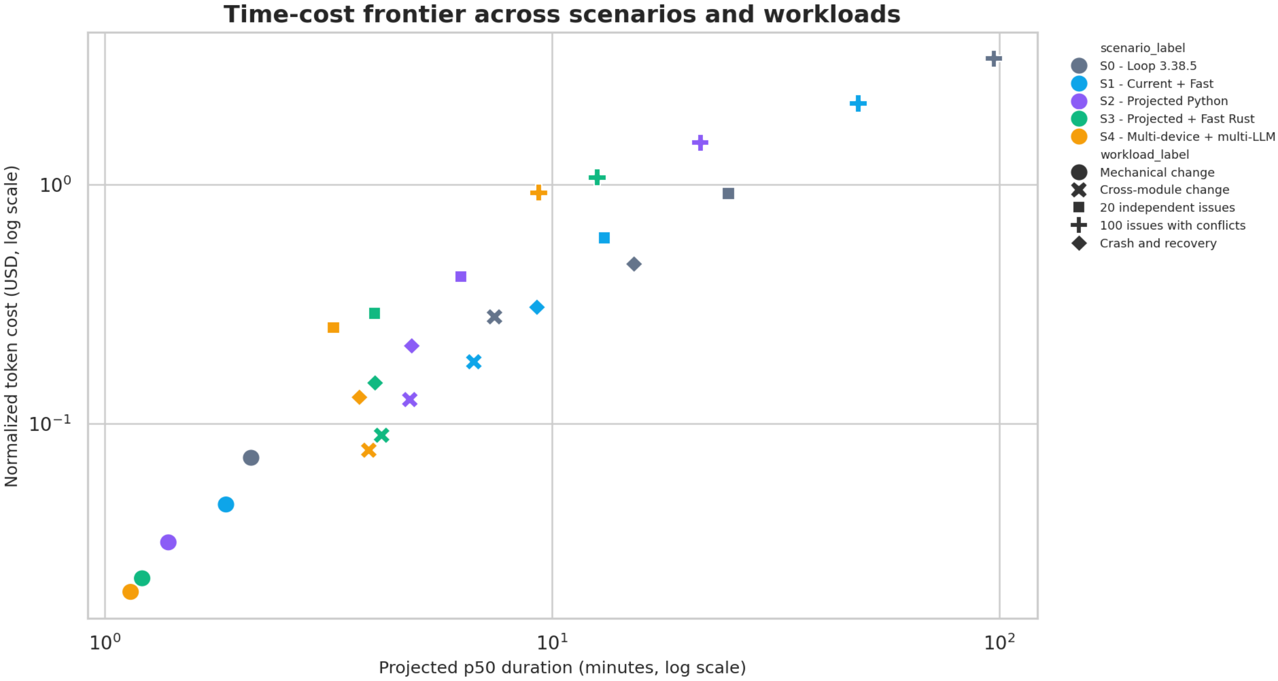
Token reduction is relative to the simulated S0 baseline for the same workload.

03 Throughput Line



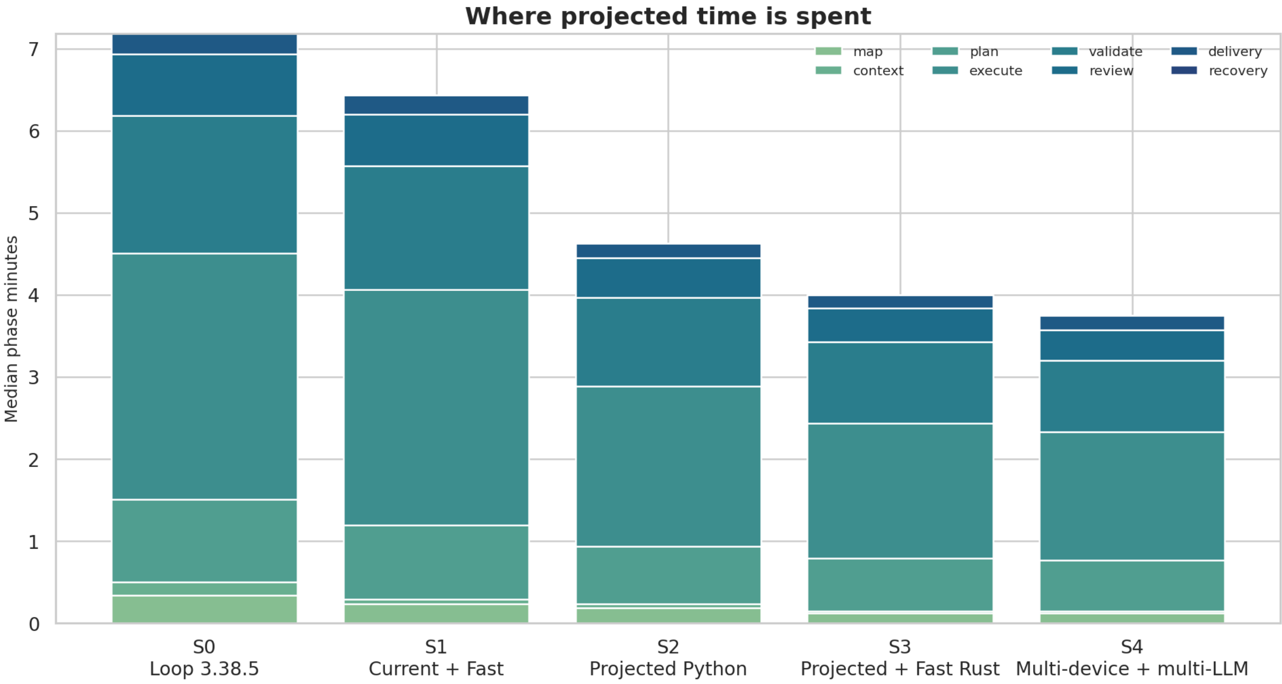
Scaling is capacity-bounded and penalized by the conflict ratio of the 100-issue workload.

04 Cost Duration Scatter



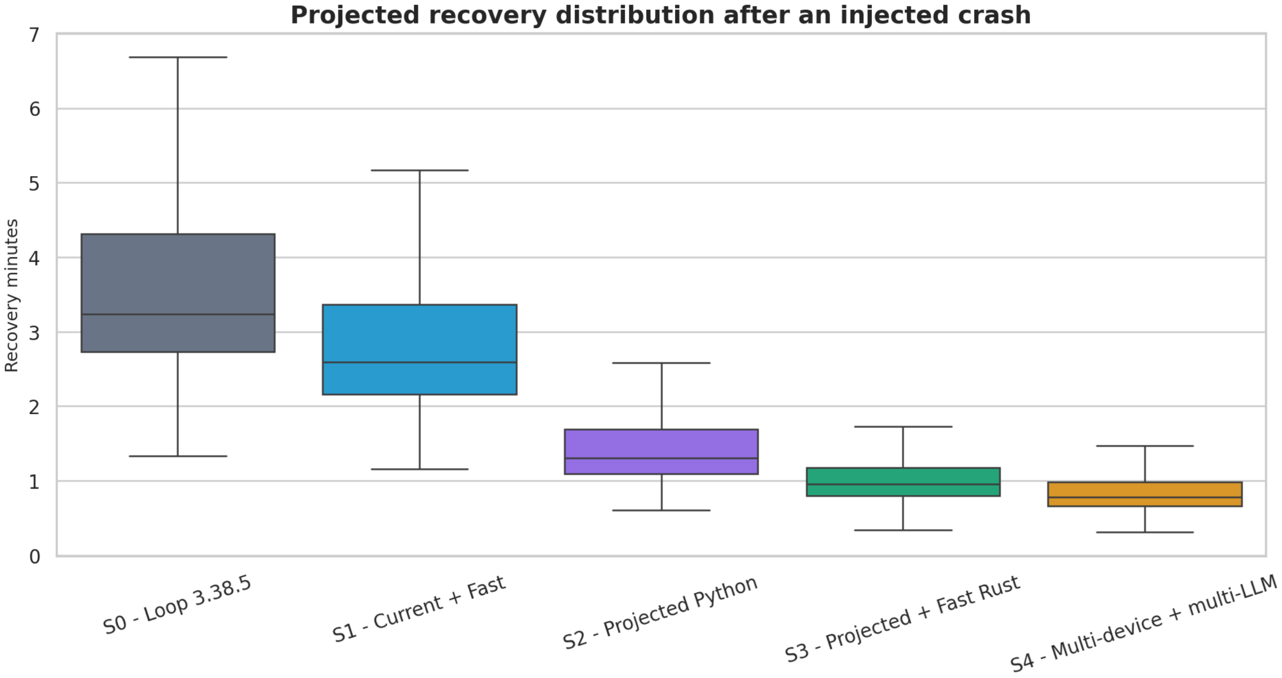
Cost uses an editable blended token-rate normalization and is not a provider price quote.

05 Phase Mix Stacked



Phase composition shows where optimization remains valuable after context and scheduling improvements.

06 Recovery Box



Recovery includes checkpoint replay plus retry work after an injected crash.

Method and assumptions

The simulator executes a deterministic Monte Carlo model. It samples lognormal phase duration and token volume, then applies queue jitter, a workload success probability and retry overhead.

Parallelism is the lower of task count and scenario capacity, reduced by a conflict penalty. Only execute, validate and review phases receive that parallel speedup.

S0 through S4 are architectural scenarios. S4 represents the projected combination of a resident Fast layer, Python control plane, Rust hot paths, bounded multi-device workers and heterogeneous runtime routing.

The simulation does not model every repository topology, model provider, rate limit, CI queue or human approval. Real receipts must progressively replace each assumed distribution.

Scenario	Tokens	LLM calls	Slots	Success delta
S0 - Loop 3.38.5	1.00x	1.00x	1	+0 pts
S1 - Current + Fast	0.65x	0.75x	2	+1 pts
S2 - Projected Python	0.45x	0.55x	4	+4 pts
S3 - Projected + Fast Rust	0.32x	0.45x	8	+6 pts
S4 - Multi-device + multi-LLM	0.28x	0.40x	16	+7 pts

Validation plan

- Replace S0 phase assumptions with receipts from a clean 3.38.5 artifact.

- Measure S1 on the same hardware, repository, provider, model and task inputs.

- Recalibrate each phase after the corresponding #801 child issue lands.

- Run at least 10 measured repetitions per lane; keep warmup separate.

- Report p50, p95, dispersion, quality and acceptance-criteria coverage together.

- Publish null plus a reason when tokens, cost, CPU, RSS or I/O cannot be observed.

- Never promote a simulated value into README or release claims.

Reproduction command

```
python3 run_projection.py --assumptions assumptions.json --output output
```

Sources and classification

- simplicio-loop pyproject.toml:
https://github.com/wesleysimplicio/simplicio-loop/blob/main/pyproject.toml

- simplicio-loop CHANGELOG.md:
https://github.com/wesleysimplicio/simplicio-loop/blob/main/CHANGELOG.md

- Projection epic #801: https://github.com/wesleysimplicio/simplicio-loop/issues/801

- Submodule PR #817: https://github.com/wesleysimplicio/simplicio-loop/pull/817

Final classification: this report is SIMULATED. It is a reusable planning and calibration artifact, not proof that version 4.0 has delivered the projected gains.